

Data Description for Tutorial 4 and 5

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Low Carbon London Trial

Following the Climate Change Act of 2008 and the associated commitment of the UK to reduce greenhouse gas emissions by 80% by 2050, the UK is poised for a significant transformation in how electricity is consumed and generated. Therefore, during 2011 and 2014, the UK government decided to conduct a series of studies called the “Low Carbon London Trial”. It is a pioneering project that has tried and demonstrated a broad range of smarter potential approaches to how distribution network operators can invest and operate in the future.

One such trial is a field experiment that tested the demand-side responses to dynamic pricing of electricity. Residential customers, facilitated through smart meters, were offered a dynamic time-of-use (dTOU) tariff, not previously tested in the UK. The values of the price bands were: high price: 67.20 pence/kWh; mid-price: 11.76 pence/kWh; and low price: 3.99 pence/kWh. To study the effects of dynamic pricing on the electricity conservation of residential customers, the researchers designed a field experiment.

The field experiment consisted of two groups, a dTOU tariff group receiving the experimental dTOU tariff (treated), and a non-Time-Of-Use (non-TOU) group receiving a standard flat tariff (control). With access to smart meters, the researcher was able to record detailed electricity usage data (at half-hour intervals), the main

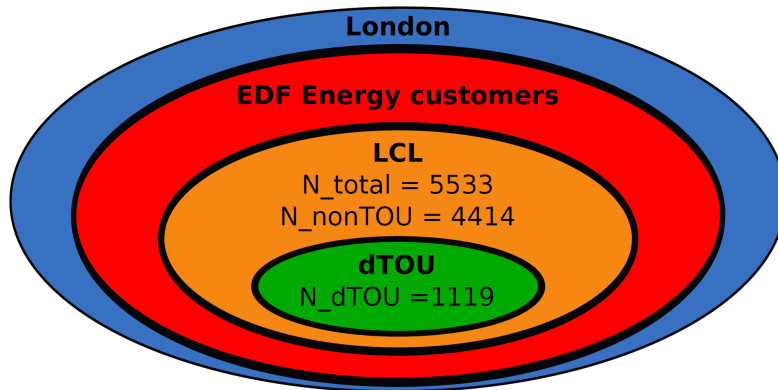


Figure 1: Venn Diagram of Sample Assignments

outcome variable of the field experiment. In addition, the experiment started on January 1, 2014 and lasted until the end of 2014. In general, we have electricity usage data for 2 years (2013 and 2014), 1 year before treatment, and 1 year after. 5,000 households were recruited in the experiment.

However, the assignment to the treated and control groups is not random. Participation in the dTOU tariff requires the formal consent of the households. In addition, households received 100 for their participation in the dTOU tariff. The following Venn diagram (Figure 1) shows the sample selection. Although the control group is a random sample of residential customers, the treatment group is based on self-selection.

In the practical session, you are given a random sample of the field experiment data. The key objective is to use the difference-in-difference method to estimate the average effect of treatment on the treated (ATT) and evaluate the validity of the DID method with the pretrend analysis.

Optimizing Bidding Strategies for Everyday Medical

Elsa (MBA at RSM '20) had just finished her second phone call with Everyday Medical (EDM), her new pricing optimization client. The issue at hand was one

she knew intimately: how to help EDM, a provider of home and durable medical equipment and supplies, optimize its online bidding strategies for Google search ads. A former engineer and computer science specialist, Elsa had recently founded Real Optimal, LLC, a startup consulting firm focused on online auction platforms. In this capacity, he was responsible for the development of bidding strategies for Google search ads.

In February 2002, Google introduced AdWords, which shaped the Internet pay-per-click advertising industry. Google AdWords was a quick and simple way to advertise on Google and its ad partners, regardless of one's budget. AdWords ads were displayed along with search results on Google, as well as on search and content sites in the growing Google Network, which included websites such as AOL, EarthLink, HowStuffWorks, and Blogger. With AdWords, users were able to associate keywords with their ads for search results on Google and its search partners, and also specify the maximum amount of money they would be willing to pay for each click (users paid only when someone clicked on their ads).

The overall pricing model of Google AdWords is based on auctions, to be more precise, generalized second-price sealed-bid auctions. The key innovations of the AdWords auctions were the introduction of Quality Scores. Quality Scores and the bidding prices (cost-per-click) codetermine the AdRank (the rank of ads) of a company. In addition, companies did not pay the full amount of a bid; instead they paid one penny more than the bid below (or, if the bidder's bid was the lowest, she paid the minimum).

Each month, EDM budgeted \$10,000 to spend on AdWords, representing 25% to 30% of monthly revenue. The company drove two-thirds of its revenues from manual wheelchairs, rolling walkers, and various other medical products and aids for daily living needed by the disabled and elderly. The average order value for such items was \$100, and nearly 90% of sales were driven by PPC advertising. The company had

historically taken a “casting a wide net” approach: bidding for the top positions in broad keywords. A significant number of clicks and cost came from searchers forming search queries that included terms or phrases such as “discount wheelchairs” and “wheelchairs for rent”. However, these keywords are generally competitive choices. EDM contacted Elsa to help them understand the improvement in the return on investment of their bidding strategies.

Managerially, for a given keyword, bidding too aggressively would generate additional clicks - at a risk, however, of paying more per additional click than what EDM could expect to earn from each of these clicks. However, bidding too conservatively might have resulted in ads appearing against fewer searches and in lower positions on the search results page, correlating to fewer clicks that might have turned out to be profitable.

Elsa was faced with a difficult problem: How to evaluate the ROI of current EDM bidding strategies? Elsa’s goal was to find the optimal bid for each keyword that would produce the maximum amount of revenue for EDM within their budget constraints. To know the optimal bid, Elsa must understand the position effects of different keywords. This is because the most important issue in keyword auction is the position of the ad on the page. Because the position of an ad is the outcome of an auction, higher positions cost more for the advertiser (all else remaining equal), and hence would be justified only if they generate higher returns for the advertiser. Measurement of causal position effects are thus of critical importance to the advertiser. However, measuring causal position effects is very challenging due to the inability to generate experimental variation in position in search advertising listings. This is because position is determined through an on-line auction, with competing advertisers bidding for their advertisements to appear in the listings.

The company offered Elsa the access to their raw data from Google Analytics. For the past campaigns run by EDM, Elsa could observe, for each campaign, how the final

Google AdWords interface showing campaign performance data. The 'Avg. Pos.' column is highlighted with an orange box.

	Keyword	Campaign	Ad group	Status	Max. CPC	Impr.	Interactions	Interaction rate	Avg. cost	Cost	Avg. Pos.
Total - all account						627,673	1,506	0.24%	\$0.20	\$298.82	3.0
●				Eligible	auto: \$1.00	40,846	21 clicks	0.05% CTR	\$0.49 per click	\$10.23	4.2
●				Eligible	auto: \$1.00	16,140	15 clicks	0.09% CTR	\$0.39 per click	\$5.87	4.1
	video			Paused	auto: \$1.00	14,917	76 clicks	0.51% CTR	\$0.32 per click	\$24.60	3.8
●				Eligible	auto: \$1.00	8,612	36 clicks	0.42% CTR	\$0.48 per click	\$17.34	1.6
●				Eligible	auto: \$1.00	3,226	1 click	0.03% CTR	\$0.63 per click	\$0.63	4.1
●				Eligible	auto: \$1.00	1,640	--	--	--	\$0.00	6.2
●				Eligible	auto: \$1.00	1,331	2 clicks	0.15% CTR	\$0.42 per click	\$0.84	3.9

Figure 2: Screenshot of AdWords Information

position was determined by the final AdRank, the revenue, the actual position, among others. The below is a screenshot of the Google Analytics of EDM.

$$\text{AdRank} = \text{QualityScores} \times \text{Bids}$$